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The debate behind the numbers: analyzing discussions about crowdsourced football player market valuations

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ABSTRACT

Purpose: This study investigates how users of the online platform transfermarkt.de rationalize their estimations of football players' market value. It responds to calls for deeper understanding of the reasoning behind crowdsourced valuations, which are increasingly used in academic and professional contexts.

Design/Methodology/Approach: We employed thematic analysis of 2,550 forum posts on *transfermarkt.de*, focusing on user discussions concerning Bundesliga players during four months of the 2022–2023 season. The posts were coded and analyzed using NVivo, following a six-phase thematic framework.

Findings: Seven overarching themes and 24 codes were identified, reflecting how users justify market value estimations through a mix of performance data, comparative reasoning, speculative rumors, and subjective assessments. Discussions often involved dissent and negotiation, underscoring the co-constructed nature of crowdsourced market values.

Practical Implications: For sport managers, analysts, and agents, the findings highlight the value of monitoring fan discourse to capture emerging reputational signals not visible in performance statistics alone.

Research Contribution: The paper advances understanding of crowdsourced valuation mechanisms and complements quantitative valuation models with qualitative insight into user reasoning.

Originality/Value: This is one of the first studies to qualitatively analyze user rationales behind market values on *transfermarkt.de*, providing insights into how digital communities shape perceptions of player worth.

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Crowdsourced market valuation; Football player valuation; Digital communities; User-generated content; Transfer market; Sport analytics

Introduction

In 2023, football clubs spent $9.63 billion on transfer fees for male football players (FIFA Global Transfer Report, 2024). A transfer fee is a cash payment from one club to another to recruit a player under contract (Frick, 2007). Player transfers significantly affect club success (Pawlowski et al., 2010), making transfer decisions a central managerial concern during the biannual transfer window.

Given this impact, fans increasingly follow transfer rumors and engage in discussions

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about players' market values, estimates that often signal potential transfer fees (Franceschi et al., 2023). This growing interest has fueled websites that estimate player market values (Frick & Prockl, 2018; Müller et al., 2017), with transfermarkt.de emerging as a key platform. On transfermarkt.de, users post and debate player market values in dedicated forums. The platform's figures are now widely cited in media, real-world negotiations (Herm et al., 2014), and academic research (Ackermann & Follert, 2018).

Prior studies on transfermarkt.de have focused on the accuracy of its market value estimates (Frick & Prockl, 2018; Herm et al., 2014) and the aggregate-level performance of the "wisdom of crowds" (Peeters, 2018). More commonly, crowdsourced values from the platform are used as proxies in quantitative research models. However, scholars such as Franceschi et al. (2023) emphasize the importance of strengthening the validity of such crowdsourced estimates for market values by analyzing the users' input. Herm et al. (2014) similarly call for deeper investigation into the platform's community discussions.

This article follows these calls and examines users' postings and their rationale for market value estimations on transfermarkt.de. Through thematic analysis of users' posts, we identify key themes and explore how users justify their market value estimations. Through the dialogue and shared estimations by the users in the form of forum postings, knowledge is co-constructed, eventually leading to the online portal's market value estimations. By examining the user posts on transfermarkt.de, we make two main contributions to the field of market value estimation: (1) How do transfermarkt.de users reason for their estimated market value? (2) How does the users' reasoning correspond or contrast with indicators from data-driven, quantitative research models on market value estimations?

This analysis advances current research by unpacking the rationale behind widely used crowdsourced values, offering insight into how such reasoning reflects, or challenges, established empirical approaches.

Background

Market values and transfer fees in football

From both sporting and financial perspectives, players are among the most significant investments in professional football (Müller et al., 2017). Quantifying the monetary value of this human capital highlights clubs' competitive advantages and informs key financial decisions such as salaries, acquisitions, and transfers. Accordingly, researchers have long sought to identify the factors that explain transfer fees and use these to estimate the value of other players (Hill et al., 2025).

A distinction has to be made between transfers, transfer fees, and market values. According to FIFA's Regulations on the Status and Transfer of Players, a transfer is defined as the formal reassignment of a player's registration between clubs (FIFA, 2022). A transfer fee is the financial compensation paid to facilitate this move (Frick, 2007). A player's market value, in contrast, is an estimate of the amount a buying club would be willing to pay to acquire a player (Herm et al., 2014). Financially, transfer fees are considered investments, while market values represent perceived worth, positioning football players as valuable human assets (Morrow, 1996, 2013). The high sums paid in transfer fees underscore the importance of accurate valuation, a subject of interest for numerous stakeholders including players, agents, clubs, and leagues (Franceschi et al., 2023).

Traditionally, market values were set by clubs or sports journalists. However, growing fan interest in transfer speculation has led to the emergence of websites that offer player valuations (Frick & Prockl, 2018; Müller et al., 2017), most notably the online portal transfermarkt.de, which has become a widely used resource.

Transfermarkt.de

Transfermarkt.de is the leading website for football transfer information (Frick & Prockl, 2018). In addition to providing football news and transfer rumors, its business model includes maintaining comprehensive data on football players, teams, and leagues (Ackermann & Follert, 2018). Beyond statistics and football news, the site, particularly its German version, is renowned for its discussion forum. Anyone can sign up as a user and participate in discussions about the market value of any football player that is in the data set of the page (Ackermann & Follert, 2018). Football fans take on the role of amateur analysts. They can propose personal market value estimations based on players' current market value estimations and discuss their suggestions with other community members. Rather than averaging estimates, transfermarkt.de relies on a few selected community members, referred to as "judges" by the platform, to aggregate and finalize player valuations (Herm et al., 2014).

Transfermarkt.de operates on the assumption that fans, collectively, can estimate market values as well as or better than experts or journalists. The portal has become a main source for reporting market values in the media and is used in real market transactions and wage negotiations (Herm et al., 2014). It has also been used in academic studies, either to assess the accuracy of valuations (e.g. Frick & Prockl, 2018), to forecast results (e.g. Peeters, 2018) or most commonly, as an input variable in empirical models using market values (e.g. Franck & Nüesch, 2012; Singh & Lamba, 2019; Tunaru et al., 2005).

Unlike data-driven analytical models that rely on statistics and performance metrics to estimate market values, transfermarkt.de is primarily based on community users' discussions and the reasoning behind their estimations. Eventually, the market values displayed by transfermarkt.de are a product of a collaborative process where users co-construct estimates through dialogue and shared expertise.

Literature review

Theoretical perspectives on market value formation

Theoretical perspectives on market value in professional sports, particularly football, emphasize the difficulty of isolating individual performance and the role of both economic and social factors. Similar challenges are found in broader debates on human asset valuation, where no consensus exists on which variables should be included (Lim et al., 2010). Similarly, in professional sports, researchers attempt to identify key indicators that determine an athlete's financial value, often represented as market value (see e.g. Di Domizio et al., 2024; Franceschi et al., 2023). In team sports like football, this valuation becomes even more complex, as performance outcomes are the product of collective team efforts rather than individual contributions alone. Team production, by nature, makes it difficult to isolate and quantify an individual player's impact on overall performance (Alchian & Demsetz, 1972). Furthermore, football is a highly interactive sport in which success depends on the combination of complementary player skills, further complicating the measurement of an individual's contribution to team performance (Carmichael et al., 2000). As a result, assessing a player's exact contribution remains inherently uncertain and requires in-depth, player-specific knowledge to estimate an appropriate monetary value (Franck & Nüesch, 2012).

Despite these complexities, player transfers have a substantial impact on a club's chances of success (Pawlowski et al., 2010). Consequently, clubs seek to acquire players who are expected to enhance team performance. However, determining how much a buying club should pay for a particular player, or what factors define a player's market value, remains a challenge that researchers from various disciplines have sought to address.

Two foundational studies that have shaped our understanding of market value dynamics in professional sports are those by Rosen (1981) and Adler (1985). In *The Economics of Superstars*, Rosen (1981) examines why a small number of top performers receive a disproportionately large share of financial rewards. He argues that individuals with marginally superior observable talent generate exponentially higher earnings and market value compared to slightly less talented individuals.

Building on Rosen's (1981) work, Adler (1985) explores why superstars emerge even among individuals of similar talent. In *Stardom and Talent*, Adler (1985) shifts the focus from purely observable talent to the role of expert recognition, arguing that talent is a hidden characteristic that requires validation from knowledgeable audiences. Furthermore, Adler (1985, 2006) contends that talent alone does not fully explain the market value of individuals. Instead, he introduces the concept of stardom, asserting that a player's financial value is driven by a combination of talent and popularity.

In the context of sports management, Adler's (1985) framework is widely used to explain why certain players command significantly higher market values than others, highlighting the interplay between performance metrics, media influence, and fan perceptions. His insights are particularly relevant to the development of data-driven market value estimations, as explored in the following section.

Empirical approaches to estimating market values

While theoretical perspectives help explain the social and structural foundations of market value in football, a large body of empirical research has sought to identify which variables actually influence player valuation. Within this empirical body of research, data-driven models on football player valuation have investigated indicators of talent as well as external determinants such as popularity-related factors (Müller et al., 2017). Whereas early studies assigned popularity to the error term (Carmichael & Thomas, 1993), more recent research on football player valuation assumes that high financial values in the form of market values are composed of the players' characteristics, performance data, and/or external factors such as popularity (Franck & Nüesch, 2008; Jedelhauser et al., 2023).

In line with Adler's (1985) concept, performance factors that would reveal talent, are the main categories that can be found in most data-driven research on the market valuation of athletes (Franck & Nüesch, 2012). Some central articles on players' performance affecting market value include, e.g. the article by the sports economists Franck and Nüesch (2012). The authors attempt to identify performance measurements that significantly affect the market value of football players. Their article assesses 20 different performance indicators that influence a player's ability to increase a team's winning potential and, therefore, explain the monetary value of the athlete. In a similar vein, Wegelin et al. (2022) focus primarily on performance indicators in their economic model to compare high-performing team members in a team sports context. Using data from the National Basketball Association (NBA), their results show that in the sudden absence of high-performing team members (e.g. due to injuries), other team members maintain their efficiency. However, the team suffers from inferior overall outcomes which underscores the high importance and value of high-performing individuals in team sports. Similarly, a recent study by Rubio Martín et al. (2024) examines how football players' injuries, and the number of days they consequently miss on the pitch, impact their market values.

Beyond performance data as a factor for market value, many data-driven estimates also include measurements of popularity to explain a high(er) valuation for certain players. A recent paper that specifically examines the popularity aspect is the article by Jedelhauser et al. (2023). The authors examine "second-

tier" stars, which they define as football players who lack popularity, and compare their contribution to team performance with that of superstars. Their findings highlight that second-tier stars, football players who lack popularity, have a lower market value. However, their marginal contribution to the team's performance is at least equal to that of superstar players who generate the highest costs (Jedelhauser et al., 2023). Their study therefore emphasizes the popularity aspect as a potential influence on a player's market value. Exploring valuation discrepancies rather than direct determinants, Campa (2022) examines the discrepancy between transfer fees and estimated player market value, showing how structural factors such as player age, contract duration, and timing of the transfer systematically shape this gap. Or, adopting a distinct conceptual lens, Leifheit and Follert (2023) propose an investment-theoretical model that estimates player value based on club-specific expectations of future payment streams, incorporating factors such as salary, merchandising potential, and sporting success.

A broader overview of empirical valuation models is provided in the systematic literature review by Franceschi et al. (2023) and earlier work by Müller et al. (2017). In the systematic literature review by Franceschi et al. (2023), the authors classify the 93 different variables found in 417 articles on football player market valuation into six categories. The authors identify these categories as key influencing parameters of football players' valuation (Franceschi et al., 2023).

However, the authors also highlight the mixed results associated with many of the variables used in previous studies. One issue they identify is "the many variations in the way common variables are defined and the inventiveness of researchers to study new variables" (Franceschi et al., 2023, p. 16). Highlighting these pitfalls is considered important for improving quantitative models of football players' valuation. In a similar vein, Follert and Gleißner (2024) criticize current empirical approaches for neglecting basic principles from corporate finance, such as future income potential and risk, which they argue are essential to a theoretically grounded valuation logic. Their critique points to a more fundamental need for conceptual consistency when modeling player market values.

One aspect that Franceschi et al. (2023) suggest as a possible way to improve the field of market value estimation is "developing the literature on crowdsourced valuation methods" (2023, p. 20). Market value estimates provided by crowdsourced platforms such as transfermarkt.de are widely used in many of research articles (e.g. Franck & Nüesch, 2012; Hill et al., 2025; Jedelhauser et al., 2023; Müller et al., 2017; Rubio Martín et al., 2024; Singh & Lamba, 2019; Tunaru et al., 2005). Therefore, Franceschi et al. (2023) highlight the importance of strengthening the validity of such crowdsourced estimates. This article aligns with the call for future research by Franceschi et al. (2023) on such crowdsourced valuation models by studying the user-generated input for market valuation on the transfermarkt.de online portal.

Crowdsourced estimations for market value

As mentioned earlier, transfermarkt.de is based on the idea that users can collectively estimate market value as accurately as, or even better than, a few experts. This reflects the principle of "wisdom of crowds" (Surowiecki, 2005), which assumes that aggregated judgments from diverse, independent contributors can yield accurate results. Several studies have assessed the accuracy of transfermarkt.de's estimates, often by comparing actual transfer fees with the site's market values (Frick & Prockl, 2018). The accuracy of these estimates is "remarkable" (Müller et al., 2017, p. 612), especially given common crowdsourcing challenges such as social influence, manipulation, and inexperience (Müller et al., 2017). According

to Herm et al. (2014), transfermarkt.de addresses these issues through a selective aggregation process known as the "judge principle." Instead of averaging all user estimates, a few empowered community members, referred to as "judges", make the final decision. These judges filter, weigh, and aggregate user input, considering the justifications provided for specific market value estimates (Herm et al., 2014). They also participate in forum discussions and explain their reasoning. In addition, they elaborate on player upgrades or downgrades in a podcast, citing user posts that support their decisions (Transfermarkt.de, 2024).

Transfermarkt.de follows a hierarchical rather than a democratic approach, which would be expected in a purely crowdsourced decision-making process. The judge principle resembles a team-leader decision-making model, in which leaders give greater attention to advice quality (Hollenbeck et al., 1995). Although final decisions rest with the judges, the estimations remain rooted in crowd wisdom and differ from purely data-driven models based on statistical metrics (Müller et al., 2017). In line with Adler's (1985) superstar theory, which emphasizes the role of popularity and audience preferences in shaping market value, it is relevant to examine how fans and amateur analysts justify their assessments. A netnographic study by Detzen and Löhlein (2024) demonstrates that ongoing user debates legitimize and sustain transfermarkt.de's valuation regime by reinforcing collective belief in the platform's practices. While their focus is on the discursive maintenance of the overall regime, this study examines how users justify individual valuation suggestions, offering a more fine-grained view of user-driven valuation processes.

Method

Data collection

To investigate transfermarkt.de's player valuations, we collated and analyzed forum entries posted by users between November 2022 and March 2023. This period was selected because, on November 9, 2022, transfermarkt.de released its biannual market value update for Bundesliga players. The timeframe captures a substantial corpus of user-generated posts, offering insights into whether users found the updates reasonable or not. It also includes the winter transfer window (January 1–31, 2023), a period that typically sparks intense discussion about player market values.

To ensure relevance and analytical focus, we selected forum discussions on male 1. Bundesliga players, as this league is the most active and widely discussed on transfermarkt.de. By focusing on this demographic, we aim to capture insightful perspectives from a highly engaged community. While limiting the scope may affect generalizability, it is justified by the league's global prominence and volume of user-generated content. Including the winter transfer window in our timeframe enabled analysis of value fluctuations and related debates during a peak period. We focused on male players because transfermarkt.de does not offer a comparable discussion function for women's football.

We selected forum entries and user discussions on transfermarkt.de because it is one of the most widely used sources for player market values in sports management and economics research (Ackermann & Follert, 2018). It is also Germany's most visited sports website and a key reference in journalistic sports reporting (Smith, 2021; The Economist, 2018).

Data analysis

This study employed thematic analysis, as outlined by Braun and Clarke (2006, 2019), to identify themes in users' forum posts related to player market value assessments. Thematic analysis was particularly well-suited for this study because user-generated discussions often feature complex, multi-layered arguments (Barab et al., 2004). Additionally, it is a

recommended method for identifying patterns across large datasets (Braun et al., 2016) and has been widely applied in recent sport management research (e.g. Andersson, 2025; Chen & Heinze, 2025). Our thematic analysis focused on uncovering recurring rationales in users' market value estimates for the 1. Bundesliga players. In this context, a *theme* represents a meaningful pattern within the data that captures insights relevant to the research question (Braun & Clarke, 2006).

A total of 2,550 forum entries served as the data corpus. The data were collected by creating a PDF version of each forum discussion page to preserve the original content. NVivo was used to systematically code and analyze the data, providing an organized and transparent framework for identifying recurring themes. Beyond structuring the coding process, NVivo also facilitated the visualization of relationships between themes, enabling a more systematic refinement process. All forum posts were published in German, a language in which the authors are either native speakers or highly proficient. Only the selected quotes included in this study were translated into English. To enhance the trustworthiness of the findings, both authors participated in the coding and analysis process, comparing interpretations and resolving discrepancies through discussion (Patton, 2002). An overview of this process is presented in Figure 1.

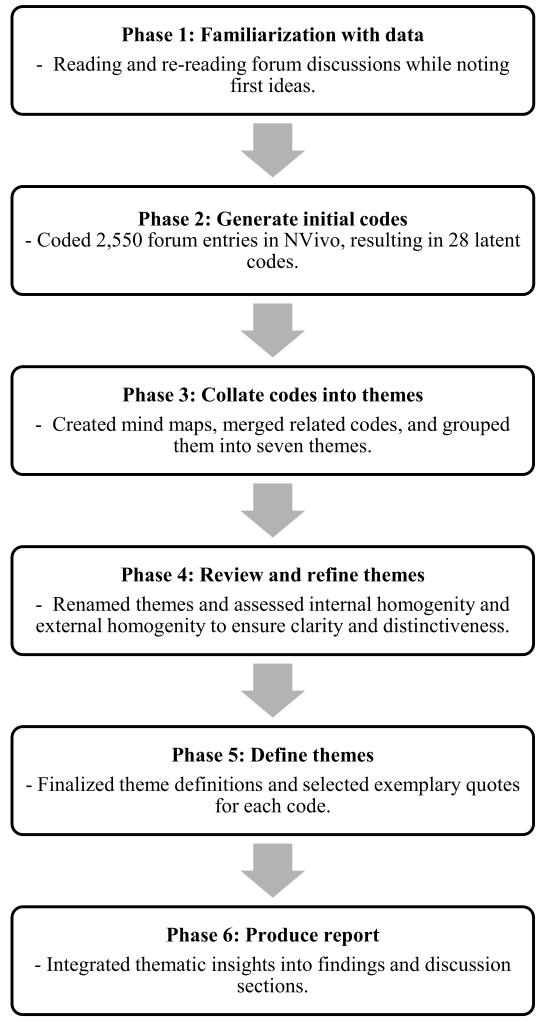


Figure 1. Coding phases.

To identify themes within the 2,550 forum posts, we applied a rigorous, time-intensive, and inductive approach guided by Braun and Clarke's (2006) six-phase framework for thematic analysis. No predefined (a priori) codes were used. Instead, coding and theme generation involved iterative cycles of revisiting and refining codes and themes. The first phase focused on familiarization with the data. While collecting the data and generating PDFs of each player's forum thread, both authors read and re-read the forum discussions, noting initial impressions. This process ensured a deep understanding of the content, which is a critical step in qualitative research (Saldana, 2021).

In the second phase, NVivo was used to systematically assign initial codes to the data. A code captures a distinct feature within the dataset that is relevant to the research objectives. As noted by Huberman and Miles (1994), coding organizes data into meaningful categories that facilitate the analysis. Data can be coded based on either semantic (explicit) or latent (underlying) features. Given the study's focus on how users construct arguments about market value, latent coding was selected.

In this phase, the main author identified 28 initial codes. Coded data differ from the unit of analysis – themes – which are broader constructs that involve interpretive analysis, developed in the third phase of thematic analysis (Braun & Clarke, 2006).

In the third phase of thematic analysis, the authors collaboratively analyzed the initial codes and collated them into overarching themes. Visual tools such as mind maps were used to organize the codes, identify relationships among them, and develop an initial set of themes. This process resulted in the merging of four codes, yielding a final set of 24 codes grouped under seven overarching themes.

In the fourth phase, the themes were reviewed and refined to ensure internal homogeneity and external heterogeneity, in line with established qualitative standards (Patton, 2002). This step was conducted collaboratively through iterative discussion and joint evaluation of the thematic structure. Several themes and a few codes were renamed to enhance clarity and to strengthen external heterogeneity. This process ensured that each theme was conceptually coherent and meaningfully from the others (Braun & Clarke, 2023). In phase five, the themes were defined following Braun and Clarke (2006). The main author carried out this step by writing brief descriptions of each theme, linking them to their corresponding codes and selecting illustrative quotes from the dataset. This phase confirmed that no further refinement was necessary. In the sixth and final phase, the findings and discussion sections were developed by integrating the thematic insights into the broader research narrative.

Findings

The analysis of user discussions resulted in seven overarching themes and 24 distinct codes. These themes capture the recurring patterns of reasoning users employ when justifying changes to a player's market value. Table 1 provides an overview of the themes and codes. The quotes referred to in the findings section are drawn from separate discussions about different players and do not represent a continuous conversation thread.

Grounding their assessments in observable and player-specific traits, users frequently refer to individual characteristics when justifying market valuations. The first theme, *Player attributes*, captures such arguments and consists of five codes: *Past or susceptibility to injuries*, *Age*, *Tactical or positional savviness*, *Skills and shortcomings*, and *Importance for the team*.

Users commonly reference *Past or susceptibility to injuries*:

> Hernandez is a super player […] but his body is simply not made to play soccer for a long time at a stretch. […] I think that €40 million due to

Table 1. Themes and codes.

Theme	Codes
Player attributes	Past or susceptibility to injuries
	Age
	Tactical or positional savviness
	Skills and shortcomings
	Importance for team
Comparative evaluation of …	… market value from another player
	… performance (subjective) to another player
	… performance data to another player
	… playing time/appearances to another player
Individual performance	Form curve
	Performance data
	User's subjective evaluation of performance
	Performance in previous seasons
International experience & exposure	Performance/appearance in international club tournaments
	Performance/(non-)nomination in the national team
Club role & playing time	Playing time
	Starting eleven
Transfer market dynamics	Contract duration
	Salary (expectations)
	Interest from another team
	(Potential) Transfer sum
	Transfer fee paid
	(Previous) Team influencing market value
User credibility disputes	Questioning the competency of other users

> the new serious injury, which will probably cost him a whole season of his career, is currently the absolute maximum what Hernandez can be worth.

When discussing *Age*, users highlight the reduced future potential of older players:

> Here you also must consider Gnabry's age by now. He was born in '95 and will be 28 in the summer. There is no longer any market value fantasy, there are no longer any talent premiums paid at that age.

Tactical or positional savviness is cited as a valuable asset:

> Hincapie has already played as CB [center back], LCB [left center back], LB [left back], and left winger at Leverkusen. He has impressed everywhere, and you rarely find such a polyvalence.

In contrast, users also emphasize *Skills and shortcomings*:

> He can't shield or forward balls. He looks very ponderous, doesn't have good top speed, and too often has bad timing.

Finally, some users underscore a player's *Importance for the team*:

> In addition, he is the undisputed captain and has played a key role in pushing a second division team to greater heights. Werder wouldn't be where they are without him.

Beyond discussing a player's individual characteristics, users frequently contextualized their assessments by drawing on comparisons, providing a broader reference framework for justifying market valuations. This reflects not only personal judgment but also alignment with transfermarkt.de's own guidelines, which encourage comparison-based reasoning. The second theme, *Comparative evaluation*, captures this pattern of argumentation, summarizing how users draw on different types of comparisons to support their positions.

The theme *Comparative evaluation …* includes the following four codes:

… of market value from another player:

> A comparable player (Ullrich, Hertha BSC II) was recently upgraded to €750k. I see Zehnter at a similar level.

… of performance (subjective) to another player:

> For me, Kimmich doesn't embody the classic leader like Ballack or Schweinsteiger once did. If I had to compare Ballack and Kimmich, I would almost see it as an insult. […] Yes, Kimmich is still a very good footballer, but €80 million is too much.

… of performance data to another player:

> With the ball, Upamecano is the more important player [than de Ligt]. Upamecano has 96 ball actions per game compared to de Ligt's 72. Long balls 5.2–3.2; passes in the opponent's half 28–20, both with similar passing rates.

… of playing time/ appearances to another player:

> Let's compare the minutes played by the two. Huseinbasic has only 723 out of a possible 2478 minutes. Engels 874 out of a possible 966 minutes.

Together with *Player attributes*, this second theme demonstrates how users not only evaluate players in isolation but also in relation to others, embedding their reasoning in broader peer contexts and collective benchmarks.

In contrast to the second theme, *Comparative evaluation*, which is grounded in relational arguments, the third theme, *Individual performance*, centers on evaluations of a single player in isolation. While comparison is excluded here by definition, the focus instead lies on player-specific justifications, demonstrating how users anchor market value suggestions in perceived individual merit rather than relative standing. The *Individual performance* theme encompasses several recurring types of arguments, captured in the following codes: *Performance data*:

> Six goals in the second half of the season for a team that only scored 12 goals in the second half of the season speaks for itself. Bülter's performance should be recognized with a higher

> market value. He has earned it in terms of performance!

Form curve:

> I don't understand the people who demand an upgrade for Kimmich in his current form. […] The player has been a shadow of his former self for a long time and is currently far away from his own claim to be a leader and a mental player.

User's subjective evaluation of performance:

> As I said, this is my sober, subjective opinion. Kimmich was never really world-class, which could justify such a market value.

And *Performance in previous seasons*:

> Poulsen still benefits massively and too much from the 2018/19 season, when he scored 15 Bundesliga goals and increased his market value to €35 million at the time.

In combination with the preceding themes, *Individual performance* further illustrates the multifaceted reasoning users employ, demonstrating how arguments can range from longitudinal assessments of past achievements to emotional, present-day evaluations of form and skill.

The fourth theme, *International experience & exposure*, captures how users factor in a player's performance and visibility on the international stage when discussing market valuations. This theme extends the reasoning beyond domestic leagues and individual form, reflecting users' assumptions that international recognition significantly enhances a player's perceived value.

Two distinct codes emerged within this theme: *Performance/appearance in international club tournaments*:

> A goal like that against Chelsea in the knockout phase of the CL [Champions League] has a huge impact on the market value.

And *Performance or (non-)nomination in the national team*:

> Accordingly, I am also in favor of a slight "adjustment", also because Sommer is still nominated for the Swiss national team (and the competition there is extremely strong).

These arguments indicate that both participation and success in globally recognized competitions, such as the UEFA Champions League or national team matches, are regarded by users as important valuation factors, reinforcing the idea that external exposure and reputation carry market relevance alongside performance.

Building on the emphasis users place on tangible indicators of performance and recognition, the fifth theme, *Club role & playing Time*, centers on players' involvement at their respective clubs. Users frequently refer to playing time and starting status as key arguments in evaluating market value. Limited appearances or the loss of a regular spot in the starting eleven are commonly cited as justification for a lower valuation. This theme was distinguished by two specific codes: *Playing time:*

> In what world should a player who has only played one game over 90 minutes, who has spent more minutes on the bench than on the pitch, and who has generally not been able to establish himself as a regular player at the 16th-placed team in the table, have a market value of €7.5 million?

And *Starting eleven*:

> Müller has lost his starting position, which should go hand in hand with a loss of status and value.

Expanding the scope beyond on-field performance, the sixth theme, *Transfer market dynamics*, captures users' arguments related to financial and contractual factors that influence a player's market value. This theme includes six distinct codes, each reflecting a different aspect of the transfer market. The *Contract duration* code, for instance, is illustrated by the comment:

> VfB has made no attempt to extend the contract, even though it ends in the summer.

Discussions around *Salary (expectations)* emphasize a player's demand and cost:

> With a rumored €6 million salary […] it is relatively clear that not only BVB (who would never pay this much salary if they didn't have to) had a strong interest and Moukoko is very much in demand internationally.

Users also highlight *Interest from another team*:

> Interest from Manchester United is known and the first figures rumored are €40 million.

The (Potential) Transfer sum is used as a point of reference for value:

> Apparently, last winter there was a €10 million offer for him from Marseille (which Leverkusen turned down). But this shows that his market value currently is not too wrong.

Comments about the *Transfer fee paid* are similarly value-oriented:

> You can now see very clearly why BVB paid over €30 million.

Finally, users sometimes argue that a player's (previous) team influences how value is perceived:

> It's unfair that so much is expected of him and that he has to score a goal every game to increase his market value just because he plays for Bayern Munich.

The final theme, *User credibility disputes*, captures discussions that do not directly assess a player's market value but instead focus on challenging the credibility, objectivity, or competence of other users – or, occasionally, the platform's evaluators. These arguments were coded separately, as they do not provide substantive input for market value estimations. Rather, they reflect a meta-level discourse in which users dispute each other's motives or analytical rigor, often by pointing out perceived bias or irrelevance. One such example reads:

> It's always interesting to read your comments. Every Bayern player, no matter how good or bad his form is, should be upgraded. But a Dortmund player, who is regarded as the world's greatest center-forward talent and who has taken a big step forward this season, should not be upgraded. […] What world do you live in? Why do you have to show your hatred for BVB and your love for Bayern in EVERY comment?

Discussion

Consistent with Franceschi et al.'s (2023) systematic review, many of the identified codes align with established quantitative determinants of football players' market value, including age, performance data, playing time, and participation in international club tournaments. Each code featured arguments that could justify both upward and downward adjustments, and none were consistently associated with only one direction of valuation change. Factors such as age or performance were invoked to support both upgrades and downgrades, depending on the context. While Frick (2007) suggests that observable factors account for a substantial portion of player valuations, our findings support this view, as user discussions predominantly focus on observable attributes such as performance metrics and player characteristics. Still, the rationales put forth by users often go beyond merely interpreting observable inputs. This analysis underscores the value of examining user-generated justifications and the co-construction of market values in online communities, contributing to the literature for four key reasons.

First, our findings underscore the dynamic interplay between performance indicators and subjective assessments, a complexity often overlooked in traditional valuation models. While data-driven models emphasize measurable performance statistics (Franck & Nüesch, 2012; Müller et al., 2017), discussions on transfermarkt.de show how users contextualize these statistics through qualitative comparisons and historical trajectories. Rather than treating objective factors such as performance data as absolute indicators that either positively or negatively affect a player's market value, users engage in comparative reasoning, situating

these factors within broader contextual elements. For example, age is evaluated not in isolation but in relation to a player's position or injury history. Similarly, in assessing *Performance/nomination to the national team*, users do not merely count a player's appearances (as quantitative models often do), but debate the relevance of (non-)nomination by considering factors such as national team strength and available alternatives. These discussions frequently evolve into debates about the appropriateness of such comparisons. This illustrates that statistical data shape player valuation, yet are equally influenced by interpretive and context-dependent reasoning.

Second, our analysis of user discussions underscores the prominence of the *Form curve* and the *User's subjective evaluation of performance* codes, revealing how recent performances strongly shape market value perceptions. Users frequently anchor their assessments to outcomes from very recent matchdays, indicating a focus on short-term fluctuations rather than longer-term trends or season-wide consistency. This suggests that crowdsourced valuations reflect a rapid feedback loop in which immediate outcomes disproportionately shape perceived player worth. Theoretically, this finding contrasts with Adler's (1985) view that superstar status emerges through accumulated recognition over time, indicating instead that the collective assessment by digital communities are more reactive and sensitive to temporal volatility.

Our third main finding concerns the theme *Transfer market dynamics* and explores how market-related factors influence player valuation. Users frequently reference market signals or rumors, such as reported (dis-)interest from other clubs, potential transfer fees, or the financial expectations of a player's current team. These discussions indicate that speculation and leaked information play a direct role in shaping perceived market value. In this context, some market factors resemble popularity effects, aligning with Adler's (1985) concept of "superstar" economics, where visibility and attention contribute to valuation.

Notably, user discussions on transfermarkt.de not only incorporate market rumors into their assessment, they also critically evaluate the reliability of sources and the plausibility of speculations. This dynamic highlights the difficulty of integrating speculative market factors into quantitative valuation models. While traditional models typically rely on measurable performance indicators, our findings suggest that user discussions may function as a barometer for market sentiment, revealing how speculation influences a player's market value. This observation aligns with research by Jedelhauser et al. (2023), who argue that "second-tier" stars, players lacking strong media-driven visibility, are often undervalued despite significant on-field contributions. Our findings build on this by showing that user engagement with, or silence around, specific market rumors can reinforce or challenge players' perceived market value.

For researchers developing quantitative market valuation models, a key challenge lies in systematically capturing the role of speculation. As directly measuring the credibility of rumors is inherently difficult, potential proxy variables could be introduced. These might include the frequency and sentiment of transfer-related discussions in online communities, betting odds for potential transfers, or media coverage of transfer speculations.

The fourth main finding of our analysis concerns the role of dissent in driving discussions. While some exchanges devolve into questioning the competency of other users, many debates are characterized by passionate and reasoned argumentation, with users presenting, challenging, and refining market value estimations. Users frequently introduce new aspects to contest others' reasoning, resulting in a highly dynamic and multifaceted discourse. Our thematic analysis shows that individual posts often include multiple rationales, underscoring the complexity of these discussions.

This finding also reinforces our first insight regarding the relational nature of indicators; users actively debate which factors should be prioritized or weighted differently based on contextual elements such as player position, team dynamics, or recent performances.

These patterns of debate, comparison, and forward-looking reasoning on transfermarkt.de reflect broader challenges in professional decision-making. From a sport management perspective, user discussions demonstrate how individuals evaluate players by combining past performance data with subjective assessments, similar to how club managers make transfer decisions. Users, like professionals, must estimate future value under uncertainty. This parallel highlights the relevance of transfermarkt.de beyond fan discourse and supports Follert and Gleißner's (2024) argument that transfer decisions often rely on subjective judgment in contexts of limited information.

Conclusion

This study responds to a call for more research on crowdsourced market valuation (Franceschi et al., 2023). By exploring the rationales behind crowdsourced market valuations, this study uncovers the similarities and differences in how users co-construct market values in comparison to data-driven approaches. Through a thematic analysis of our data, we identified seven main themes and 24 codes illustrating how users rationalize their market value estimates. In particular, the relational nature of valuation factors and the role of dissent-driven discussions highlight the unique value of transfermarkt.de's platform, as well as the added benefits, and potential drawbacks, of crowdsourced valuations compared to empirical market value estimation. In a crowdsourced setting, users co-construct market value estimates through lively debates. Reaching consensus ultimately requires a final judgment by an authority or team leader. In the case of transfermarkt.de, this authority is the community judges, who finalize market value updates. However, our findings suggest that these decisions do not necessarily resolve valuation debates but instead sustain ongoing discussions, reinforcing the iterative nature of knowledge co-construction.

Our study demonstrates that while empirical models rely on historical performance data and statistical benchmarks, fan-driven discussions integrate contextual, relational, and speculative dimensions that are often absent from empirical valuation frameworks. This aligns with Adler's (1985) argument that market value is shaped not only by talent but also by social perception and media influence. By revealing how users incorporate both objective metrics and subjective interpretations, our findings contribute to a more comprehensive understanding of player valuation, particularly the understanding of market values as constructed on transfermarkt.de, which are frequently used in both academic research and professional practice.

The findings of this study have important implications for practice, particularly for stakeholders in the sports industry, such as analysts, scouts, and clubs. Recognizing the value of crowdsourced market value estimations from platforms like transfermarkt.de, practitioners can gain a more nuanced perspective, taking into account contextual and comparative factors that extend observable past variables. For clubs and agents, this means that monitoring fan discourse could provide additional market intelligence. Similarly, for researchers, incorporating elements of user reasoning into market value models could refine predictive accuracy, bridging the gap between quantitative analysis and crowdsourced valuation processes.

Our study also underscores the role of digital communities as influential actors in market valuation. As the sports industry continues to embrace data-driven decision-making, acknowledging the interplay between empirical models and fan-driven valuations will be essential in capturing the full complexity of player valuation in professional football.

Disclosure statement

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